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QTopic: A novel quantum perspective on learning topics from text

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ABSTRACT

Topic modeling is an unsupervised technique in natural language processing (NLP) used to identify hidden topic structures within large text datasets. Among traditional approaches to topic modeling, latent Dirichlet allocation, BERTopic, and Top2Vec, are widely adopted to uncover hidden topics in text data. However, these methods often struggle with poor performance in scenarios involving limited data availability or high-dimensional textual features. In this research, we propose QTopic, a novel hybrid quantum-classical topic modeling architecture that leverages quantum properties through parameterized quantum circuits. By integrating quantum-enhanced sampling into the inference pipeline, the proposed model captures richer topic distributions by mapping textual data into a higher-dimensional space. Benchmark experiments demonstrate that QTopic consistently outperforms classical approaches in terms of coherence, diversity, and topic distinctiveness, particularly when modeling a small number of topics. This study demonstrates the promise of quantum techniques in advancing unsupervised NLP, while also highlighting hardware limitations that present challenges for future research.

1. Introduction

With the exponential growth of textual data across domains such as news, research, and social media, extracting meaningful information from large document collections has become an increasingly challenging task. Topic modeling facilitates the extraction of meaningful themes in large text datasets through unsupervised analysis. By uncovering patterns in word usage and grouping related terms, it provides a deeper insight into document content and supports activities, for example, information retrieval [21], document classification [53], and summarization [13].

Various classical and modern methods are utilized for topic modeling to better understand and analyze documents. Latent Dirichlet allocation (LDA) is a popular and widely used statistical method that models topics as probability distributions over words [6,23]. This approach has found broad applications in multiple fields, such as identifying research trends [20,54], analyzing consumer complaints [3], and extracting themes from journal abstracts [2]. Despite its effectiveness, this approach requires extensive hyperparameter tuning and a large corpus to identify meaningful topics effectively. Deep learning–based approaches, such as BERTopic, have gained significant attention recently by leveraging transformer embeddings for more semantically meaningful topics [19,40]. Although this pre-trained approach effectively identifies topics in complex datasets, its computational cost scales substantially with larger corpora. Similarly, Top2Vec uses vector embeddings and clustering techniques to discover topics directly from document representations without requiring predefined topic numbers [1]. However, these traditional models often struggle to capture complex dependencies, topic correlations, and perform poorly in short texts due to simplified assumptions and approximate inference [18,22]. As the demand for more expressive and interpretable topic models increases, there is a growing interest in exploring novel computational paradigms beyond classical probabilistic frameworks.

In recent years, quantum machine learning has garnered growing interest from the research community, particularly in domains such as image processing [24,38,50], and natural language processing [15,46]. By leveraging foundational quantum principles, including superposition, entanglement, and interference, quantum algorithms are capable of encoding and manipulating data in exponentially large Hilbert spaces. This high-dimensional representation enables richer and more expressive modeling of complex structures in data, particularly for tasks involving unstructured or sparse inputs [11,27]. For textual data, these quantum representations facilitate novel ways of capturing semantic relationships that may be difficult to model with classical approaches. Motivated by these advantages, recent studies have begun exploring the integration of quantum computing into topic modeling. The goal is to improve the discovery of latent semantic structures in large document corpora, potentially enhancing topic coherence, separation, and diversity beyond

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what is achievable with classical methods [39]. This emerging line of research seeks to bridge quantum computing with natural language understanding, offering new opportunities for scalable and interpretable topic discovery.

In this research, we propose QTopic, a quantum-enhanced topic modeling framework that integrates quantum circuit inference into the topic discovery process. QTopic follows the standard topic modeling pipeline but replaces the variational sampling step with quantum circuit inference using encoded document embeddings, enabling richer topic correlations through entanglement and measurement. This hybrid quantum-classical architecture makes it more practical for current quantum hardware while maintaining scalability [51]. This approach not only enhances topic coherence and diversity but also introduces a new pathway for embedding quantum advantage into interpretable topic models.

The key contributions of this study include:

i. We present QTopic, a novel hybrid quantum-classical topic modeling framework that leverages parameterized quantum circuits, offering a new perspective beyond classical probabilistic models.
ii. We integrate classical document preprocessing with quantum topic inference, leveraging quantum properties to enable richer and more expressive representations of latent topics.
iii. We designed the model to scale with the number of topics by mapping each topic to a quantum qubit, allowing experimentation with increasing topic counts within quantum hardware constraints.
iv. We conduct comprehensive benchmarking of QTopic against state-of-the-art models, including classical LDA, BERTopic, and Top2Vec, across multiple configurations (4, 8, and 16 topics). QTopic demonstrates competitive or superior performance in generating higher-quality and more interpretable topics, especially at lower topic counts.

The remaining chapters of this paper are organized as follows. Section 2 reviews related studies in classical topic modeling and quantum natural language processing (QNLP). Section 3 presents the proposed methodology, including feature preparation, quantum topic inference, and topic assignment. Section 4 outlines the experimental design and implementation details, followed by the presentation and analysis of results in Section 5. Finally, key findings, challenges, limitations, and future research directions are summarized in Section 6.

2. Related work

Topic modeling aims to uncover latent semantic structures from large text corpora by representing documents as probabilistic mixtures of hidden topics. Over the past two decades, various classical, neural, and recently quantum-inspired models have been proposed to improve topic coherence, interpretability, and computational efficiency. This section reviews the key developments relevant to the proposed QTopic framework, focusing on classical models, deep learning models, and quantum approaches for natural language processing.

2.1. Classical topic modeling

Classical topic modeling has been a fundamental approach for discovering latent semantic patterns within large text corpora. The widely used probabilistic model LDA [6] represents a document as a mixture of hidden topics, with each topic being a probability distribution over words. The underlying generative process relies on Dirichlet priors for document–topic and topic–word distributions, estimated through Gibbs sampling or variational inference. However, LDA's assumptions of independent words and its lack of contextual or hierarchical modeling reduce its effectiveness in capturing coherent topics.

To address these limitations, several extensions have been proposed. Correlated topic models (CTM) [5] capture dependencies between topics by replacing Dirichlet priors with logistic-normal distributions, while hierarchical LDA (hLDA) [17] introduces tree-structured priors to model topic hierarchies. Matrix factorization approaches such as non-negative matrix factorization (NMF) [26] have also been explored as deterministic alternatives, decomposing the document–term matrix into interpretable latent factors.

Recent studies have refined probabilistic topic modeling by introducing sparsity-inducing priors [35], as well as nonparametric extensions such as the Hierarchical Pitman–Yor and Nested Chinese Restaurant processes [34,47]. Further advances in variational and stochastic inference have improved scalability and interpretability for large-scale corpora [45]. Despite these improvements, classical models continue to struggle with contextual and long-term semantic dependencies, motivating the development of more advanced approaches.

2.2. Deep learning based topic modeling

With the advancement of neural architectures, deep learning-based topic models have emerged as powerful alternatives to traditional probabilistic approaches. Neural variational document models (NVDM) [32] introduced the variational autoencoder (VAE) framework to learn latent topic representations through neural inference networks. Word embeddings such as Word2Vec [33] and GloVe [36] have enabled semantically richer topic representations by capturing contextual similarity in continuous vector spaces.

More recently, transformer-based approaches such as Top2Vec [1] and BERTopic [19] leverage contextual embeddings from models like BERT [10] to cluster semantically related documents without explicit probabilistic assumptions. Deep embedded clustering methods [52] and graph-based neural topic models [29] have also demonstrated further improvements in topic separation and interpretability. However, implementation of these models often depends on large datasets and high computational resources, and they may lose interpretability due to the opacity of neural representations.

2.3. Quantum computing for NLP and topic modeling

Recent progress in quantum computing has motivated new approaches to NLP grounded in quantum mechanics. These quantum-inspired methods encode linguistic data as quantum states in high-dimensional Hilbert spaces, enabling more expressive and non-classical forms of probabilistic inference.

Early theoretical work on QNLP was motivated by the DisCoCat (Distributional Compositional Categorical) model [9], which unifies compositional grammar with distributional semantics and demonstrates that grammatical composition can be captured by tensor contractions. This framework was later extended into executable quantum algorithms [31], introducing quantum circuits for compositional semantics that utilize entangled states to model word dependencies and contextual relationships. These contributions laid the foundation for quantum-enhanced representations of meaning and demonstrated that linguistic structures can be naturally embedded into quantum computational spaces.

Building upon these theoretical insights, quantum machine learning techniques have been applied to a range of NLP tasks. Quantum algorithms have been shown to offer exponential representational advantages for data classification and similarity search [44]. Subsequent studies applied variational quantum circuits (VQCs) and quantum feature maps to NLP problems such as text classification, sentiment analysis, and document clustering [8,37,55]. In these hybrid quantum–classical models, quantum layers are trained jointly with classical neural networks to perform feature extraction or dimensionality reduction within the quantum Hilbert space. Such hybridization leverages quantum parallelism for encoding richer semantic relationships while maintaining classical interpretability and scalability. These advances highlight the potential of quantum-enhanced topic modeling frameworks such as QTopic, which integrate quantum sampling within classical probabilistic inference to improve topic coherence and separation.

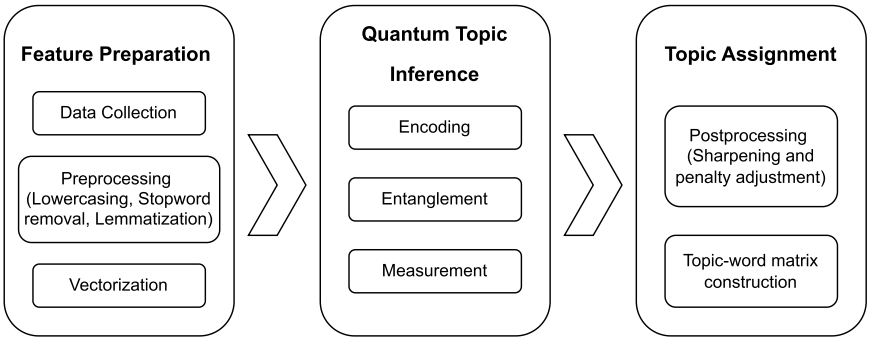


Fig. 1. Overview of the proposed QTopic workflow. The process involves three main stages: (i) Feature preparation, where documents are preprocessed and vectorized; (ii) Quantum topic inference, using angle embedding and parameterized quantum circuits to sample topic distributions; (iii) Topic assignment, where measurement outcomes are used to update document-topic associations.

3. Proposed methodology

This section introduces our proposed QTopic approach, designed to achieve more expressive and well-separated topic modeling compared to traditional methods. The QTopic framework consists of three main modules: (i) feature extraction, (ii) quantum topic inference, and (iii) topic assignment. These are demonstrated in Fig. 1. The complete sequence of steps of the proposed model is outlined in Algorithm 1, with detailed explanations provided in the following sections.

3.1. Feature preparation

The first module of the QTopic framework processes raw text into structured features suitable for quantum encoding. Each document is preprocessed through a sequence of steps, and the resulting tokens are vectorized. The vectors are transformed using the principal component analysis (PCA) dimensionality reduction technique, allowing for compatibility with the fixed qubit count of the quantum circuit.

3.1.1. Text preprocessing and normalization

Text preprocessing is a fundamental step in NLP tasks. In this work, a sequence of preprocessing operations is carried out on the corpus (D) to produce consistent and semantically meaningful input data. For each document $d \in D$, text is tokenized, lowercased, and filtered to remove stopwords and non-alphabetic tokens. To obtain meaningful word tokens and reduce words to their base forms, lemmatization is applied [49]. This helps ensure that different forms of a word are considered as an individual, meaningful token.

The resulting tokenized document can be written as follows:

$$\text{tokens}_d = \text{Lemmatize}(\text{Tokenize}(d_{\text{lower}})) \setminus \text{Stopwords} \tag{1}$$

Algorithm 1 Proposed QTopic: Quantum-Inspired Topic Modeling.

Require: Corpus $D = \{d_1, \ldots, d_N\}$, number of topics T, PCA dimension p, top-k words
Ensure: Top-k words per topic, coherence and diversity scores
Preprocess each d_i: lowercase, tokenize, remove stopwords, lemmatize
Compute TF-IDF matrix X, reduce with PCA to p dims $\rightarrow X_p$
Normalize rows of X_p, pad/truncate to $n = T$ (number of qubits)
for each document vector $x \in X_p$ **do**
  Encode x using $RY(x_i)$ on n qubits
  Apply parameterized entangling layers; measure $\langle Z_i \rangle \rightarrow z$
  Compute probabilities: $p = (z+1)/2$, then normalize over T topics
  Sharpen: $p_t \leftarrow p_t^3 / \sum p_t^3$
  Apply diversity penalty: $p_t \leftarrow p_t \cdot \exp(-n_t / \max(n_t))^2$
  Assign to topic $t^* = \arg\max p_t$, update topic count n_{t^*}
end for
for each topic t **do**
  Aggregate TF-IDF vectors of documents assigned to t
  Compute average topic-word weights and normalize
end for
Extract top-k words per topic from topic-word matrix
Compute coherence scores and topic diversity
return Top-k words per topic, coherence scores, diversity scores

3.1.2. Vectorization via TF-IDF

The preprocessed tokens are then transformed into quantitative features through term frequency–inverse document frequency (TF-IDF) [41]. The TF-IDF score for a term t in document d is computed as:

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \cdot \log\left(\frac{1 + N}{1 + \text{df}(t)}\right) \tag{2}$$

where $\text{tf}(t, d)$ is the frequency of term t in document d, $\text{df}(t)$ indicates the count of documents in which t occurs, and N is the total number of documents in the corpus. The addition of 1 in the numerator and denominator is a smoothing factor to ensure numerical stability by preventing zero-valued denominators and moderating the effect of very rare terms.

3.1.3. Dimensionality reduction using PCA

The resulting TF-IDF matrix resides in a high-dimensional feature space, making it impractical for direct use in quantum circuits due to hardware and qubit limitations. To address this, PCA is applied to retain the most informative components of TF-IDF features, capturing dominant patterns in the document corpus while reducing computational complexity [30]. Let $\mathbf{X}_{\text{tfidf}} \in \mathbb{R}^{N \times V}$ denote the TF-IDF matrix, where N represents the total count of documents and V is the size of the vocabulary. The transformation is given by:

$$\mathbf{X}_{\text{pca}} = \mathbf{X}_{\text{tfidf}} \cdot \mathbf{W}_{\text{pca}}, \quad \mathbf{W}_{\text{pca}} \in \mathbb{R}^{V \times d} \tag{3}$$

where $d \ll V$ denotes the number of principal components retained. Each resulting document vector $\mathbf{x}_d^{\text{pca}}$ is then normalized using the ℓ_2-norm:

$$\mathbf{x}_d^{\text{norm}} = \frac{\mathbf{x}_d^{\text{pca}}}{\left\|\mathbf{x}_d^{\text{pca}}\right\|_2} \tag{4}$$

The normalized vectors are subsequently padded or truncated to a fixed length $n = K$, where the value of K represents the number of topics extracted and is equivalent to the number of qubits employed in the quantum system.

3.2. Quantum topic inference

Quantum inference leverages the quantum properties to map the features into high-dimensional space to capture complex, non-linear relationships between topics. A parameterized quantum circuit encodes each document vector into a corresponding quantum state for topic discovery. The process consists of three key steps: encoding, entanglement, and measurement.

3.2.1. Encoding

Encoding transforms classical data into quantum states, as quantum circuits cannot operate directly on classical inputs [28]. Generally, classical data can be transformed into quantum states using one of three encoding techniques: basis encoding, angle encoding, or amplitude encoding [42]. In this study, we employ angle encoding to enable the quantum circuit to effectively capture and manipulate the semantic structure of the input data [7]. Specifically, each component x_i of the input document vector $\mathbf{x} \in \mathbb{R}^n$ is encoded into the quantum circuit through a rotation around the Y axis. A parameterized rotation gate $RY(x_i)$ is applied to the associated qubit q_i, effectively embedding the classical feature into the quantum state. The operation is defined mathematically as:

$$U_{\text{input}}(\mathbf{x}) = RY(x_0) \otimes RY(x_1) \otimes \cdots \otimes RY(x_{n-1}) = \bigotimes_{i=0}^{n-1} RY(x_i) \tag{5}$$

which represents the tensor product of individual rotations applied to each qubit.

3.2.2. Entanglement

Following the encoding, an entanglement operation is applied to capture complex correlations and interactions between topics. Entanglement creates dependencies between qubits, which enables the quantum circuit to represent more expressive latent topic structures than independent classical features. Let the circuit apply L layers of trainable quantum operations, which consist of a layer of RX rotations parameterized by weights $\theta_i^{(\ell)}$, followed by a ring of CNOT gates for entanglement (Fig. 2). The full parameterized circuit is thus given by:

$$U(\mathbf{x}, \boldsymbol{\theta}) = \left[\prod_{\ell=1}^{L} \left(\bigotimes_{i=0}^{n-1} RX\left(\theta_i^{(\ell)}\right) \cdot \prod_{i=0}^{n-1} \text{CNOT}_{i \to (i+1) \bmod n} \right) \right] \cdot \left(\bigotimes_{i=0}^{n-1} RY(x_i) \right) \tag{6}$$

3.2.3. Measurement

After the encoding and entanglement operations, quantum information is converted into a classical representation through measurement.

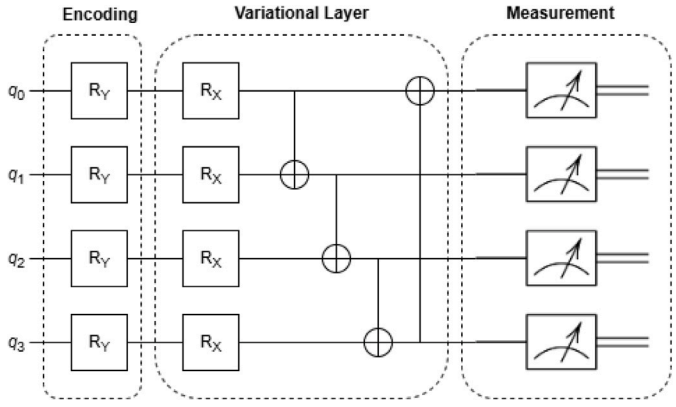


Fig. 2. Parameterized quantum circuit for topic inference in Q-Topics. The circuit applies angle embedding to encode document features into quantum states, followed by layers of parameterized rotation and entanglement gates to model topic distributions. Final measurements extract probabilistic outputs corresponding to topic assignments.

On each qubit, the circuit measures the expectation value via the Pauli-Z operator, producing a real-valued output vector corresponding to the encoded and transformed document representation.

$$\mathbf{z}_d = \left[\langle Z_0 \rangle, \langle Z_1 \rangle, \ldots, \langle Z_{K-1} \rangle \right] \tag{7}$$

The measurement results are then transformed into probabilities $\mathbf{p}_d$ by rescaling and normalization:

$$\mathbf{p}_d = \frac{\mathbf{z}_d + 1}{2}, \quad \mathbf{p}_d \leftarrow \frac{\mathbf{p}_d}{\sum_j p_d[j] + \epsilon} \tag{8}$$

Here, the index j runs over the components of the quantum output vector $\mathbf{p}_d$, corresponding to each qubit. The normalization ensures $\mathbf{p}_d$ forms a valid probability distribution. Finally, ϵ is a small constant for numerical stability.

3.3. Topic assignment

After quantum topic inference, the topic assignment step maps sampled topic indices to documents and words. Postprocessing then sharpens the distribution, applies penalty adjustments, and constructs the final topic-word matrix to enhance coherence and interpretability.

3.3.1. Postprocessing

Several postprocessing steps are applied to the output of the quantum circuit to achieve meaningful and balanced topic assignments. First, a sharpening function is applied to amplify the dominant topic probabilities while attenuating weaker ones. This enhances the confidence of the topic distribution, resulting in more decisive and interpretable topic assignments. The topic distribution vector $\mathbf{p}_d \in \mathbb{R}^K$ for document d is sharpened using a temperature parameter $\gamma > 1$:

$$\mathbf{p}_d^{\text{sharp}} = \frac{(\mathbf{p}_d)^\gamma}{\sum_j (\mathbf{p}_d[j])^\gamma} \tag{9}$$

However, confident predictions alone can lead to topic collapse, where only a few topics dominate across the corpus. To mitigate this, a diversity regularization term is introduced, which penalizes topics that have been frequently assigned by down-weighting their contribution. This exponential penalty discourages overdominance of frequently assigned topics. Here, a regularization term α_k is computed based on topic frequency, where c_k represents the number of documents assigned to topic k, and ε is a small constant to avoid division by zero.

$$\alpha_k = \exp\left(-\left(\frac{c_k}{\max_k c_k + \varepsilon} \right)^2 \right) \tag{10}$$

The final adjusted topic distribution for document d incorporates both sharpening and diversity weighting. This yields a normalized vector $\tilde{\mathbf{p}}_d \in \mathbb{R}^K$, reflecting both confidence and diversity-aware weighting.

$$\tilde{\mathbf{p}}_d = \frac{\mathbf{p}_d^{\text{sharp}} \cdot \boldsymbol{\alpha}}{\sum_k \mathbf{p}_d^{\text{sharp}}[k] \cdot \alpha_k + \varepsilon} \tag{11}$$

Each document d is then assigned to the topic with the highest adjusted probability, ensuring both interpretability and robustness in topic allocation.

$$t_d = \arg\max_k \tilde{\mathbf{p}}_d[k] \tag{12}$$

3.3.2. Topic-word matrix construction

For each topic k, we aggregate the TF-IDF vectors $\mathbf{x}_d^{\text{tfidf}}$ of all documents $d \in D_k$ assigned to that topic:

$$\mathbf{w}_k = \sum_{d \in D_k} \mathbf{x}_d^{\text{tfidf}} + \epsilon \tag{13}$$

Here, $\mathbf{w}_k \in \mathbb{R}^V$ is the raw word score vector for topic k, and V is the vocabulary size.

To reduce the influence of commonly occurring terms, we have normalized and suppressed high-frequency words. Let $\mathbf{u} \in \mathbb{R}^V$ denote the cumulative word usage across all topics.

$$\mathbf{w}'_k = \frac{\mathbf{w}_k}{\mathbf{u} + \epsilon} \tag{14}$$

Finally, the topic-word matrix $\mathbf{T} \in \mathbb{R}^{K \times V}$ is obtained by row-normalizing all adjusted topic-word vectors:

$$\mathbf{T} = \text{row-normalize}([\mathbf{w}'_0, \mathbf{w}'_1, \ldots, \mathbf{w}'_{K-1}]^\top) \tag{15}$$

Each row of $\mathbf{T}$ represents a probability distribution over words for a given topic.

4. Experimental setup and evaluation

We outline the experimental configuration employed to assess the performance of the proposed QTopic model. This includes a detailed description of the dataset, computing environment, model settings, and the evaluation metrics applied to assess topic quality.

4.1. System configuration

The experimental setup utilized a system powered by an Intel(R) Core(TM) i9-14900 processor and an NVIDIA GeForce RTX 4090 graphics card. All experiments were conducted on a Windows 11 platform, with Python 3.10 serving as the main programming language. All model implementations leveraged GPU acceleration via CUDA 11.8 and cuDNN 8.6 to enhance computational efficiency. For the quantum components, simulations were carried out using the PennyLane framework [4].

4.2. Model setup

In this experiment, we compare our proposed QTopic model with three baseline models: (i) LDA, (ii) BERTopic, and (iii) Top2Vec. To ensure a fair comparison, similar hyperparameter settings are applied across all models. We evaluated the models using different numbers of topics: 4, 8, and 16. Common configurations include the maximum number of features, selected models, maximum and minimum document frequency thresholds, and the number of iterations. For QTopic specifically, the number of qubits is set equal to the number of topics, and the quantum circuit is constructed using ten PCA components and seven layers. Full details are provided in Table 1.

4.3. Datasets

In this experiment, three publicly available datasets are employed to evaluate the performance and generalizability of the proposed model. A detailed description of these datasets, including the number of documents, average document length, and category information, is presented in Table 2.

20 Newsgroups dataset: The 20 Newsgroups dataset [25] is a widely recognized benchmark corpus commonly used to evaluate topic discovery algorithms. It comprises 11,314 training documents drawn from 20 thematic categories, such as politics, sports, science, religion, and technology. The data is originally partitioned into training and testing subsets; however, only the training subset provided by the scikit-learn library is utilized in this experiment. After standard preprocessing steps, the average document length is approximately 271 words, and the vocabulary size is about 25,000 unique tokens. The categories are relatively balanced, with each topic containing an average of 500 to 600 documents.

BBC News dataset: The BBC News dataset [16] contains 1,490 news articles categorized into five topics: business, entertainment, politics, sport, and technology. Only the training subset is utilized for unsupervised topic modeling. The average article length is around 208 words, and the vocabulary size is approximately 9,800 words. The dataset exhibits a balanced class distribution, with each category comprising nearly 300 articles.

TripAdvisor dataset: The TripAdvisor dataset consists of 20,491 user reviews related to hotels, attractions, and travel experiences. Although the original data includes metadata such as ratings and locations, only the textual content is used in this study. After preprocessing, the average review length is approximately 94 words, and the vocabulary size is around 15,200 tokens. As the dataset is unlabeled, it serves as an unsupervised corpus to evaluate the model's ability to infer latent topics.

Table 1
Summary of hyperparameters for topic modeling models.

Model	Key parameters	Features	Other settings
LDA	passes = 10	max_features = 3000	no_below, no_above = 5, 0.95
Top2Vec	speed = deep-learn	embedding = USE	workers = 4
QTopic	n_qubits = 4,8,16	pca_components = 10	n_layer = 7

Table 2
Description of datasets used for topic modeling.

Dataset	Domain	#Docs	Avg Len	#Category	Vocab size
20 Newsgroups	Online Forum	11,314	~271	20	~ 25,000
BBC News	News Articles	1,490	~208	5	~ 9,800
TripAdvisor	User Reviews	20,491	~94	/	~ 15,200

4.4. Evaluation criteria

The performance of the topic models is evaluated using three key metrics: vector-based topic coherence (C_V), topic diversity (TD), and Jaccard similarity (JS). The C_V coherence metric is selected as the primary evaluation criterion for human interpretability, as it captures both co-occurrence patterns and semantic similarity within each topic [43]. To identify the uniqueness and distinctiveness of each topic within the model, the topic diversity metric is employed. Finally, Jaccard similarity is used to quantify the degree of overlap between topic word sets, providing insights into topic redundancy within the model [48].

5. Result analysis

This section demonstrates the results of the proposed QTopic model on three benchmark datasets: 20 Newsgroups, BBC News, and TripAdvisor. The evaluation covers qualitative visualization, coherence and diversity analysis, and ablation studies to assess the model's performance in topic extraction and semantic representation.

5.1. Qualitative analysis and visualization

In this section, we present qualitative analysis and visualizations of the topics discovered by the proposed model. These include topic words with inferred categories, word clouds, topic-word probability distributions, document–topic distributions, and intertopic distance maps. Together, these keywords and visualizations provide deeper insight into the quality, distinctiveness, and interpretability of the topics identified.

To assess the quality of the topics discovered, we present the top 10 words for each of the 8 topics. These words reflect the underlying categories identified within the 20 Newsgroups, BBC News, and TripAdvisor datasets, as presented in Tables 3–5, respectively. We also visualize the top words for each topic using word clouds for the BBC News dataset (Fig. 3). The word clouds highlight the most representative keywords per topic, with font size indicating relative importance based on word probability. The visualizations illustrate that the topic keywords are distinct and semantically relevant to their respective categories.

Additionally, we present the top-5 words for each topic along with their corresponding probabilities (Fig. 4). These visualizations demonstrate that the topic words discovered by QTopic are coherent, distinct, and capture meaningful semantic structure.

Table 3
Topic words and inferred categories on 20 Newsgroup dataset (num topics = 8).

#Topic	Top words	Inferred category
0	car, bike, writes, article, like, university, distribution, usa, good, space	General Discussion / Student Life / Auto
1	geb, israeli, msg, arab, surrender, nazi, shameful, jake, chastity, skepticism	Middle East Politics and Ideology
2	armenian, armenia, serdar, argic, azeri, henrik, scsi, serum, turkish, azerbaijan	Caucasus Geopolitics
3	vga, ati, vesa, cursor, svga, icon, ultra, meg, nubus, vlb	Display and Graphics Hardware
4	ide, scsi, simms, simm, motherboard, fpu, floppy, dma, centris, quadra	Computer Hardware and Architecture
5	encryption, crypto, escrow, wiretap, nsa, bontchev, amanda, sternlight, encrypted, rsa	Cryptography and Surveillance
6	scripture, gospel, atheism, verse, arrogance, revelation, belief, commandment, worship, doctrine	Religion and Belief Systems
7	espn, playoff, nhl, hockey, bruin, cub, lindros, lemieux, pitching, detroit	Sports (Hockey and Baseball)

Table 4
Topic words and inferred categories for BBC News dataset (num topics = 8).

#Topic	Top words	Inferred category
0	club, said, chelsea, united, game, arsenal, league, sale, player, goal	Football / Premier League
1	roddick, indoor, holmes, williams, tennis, marathon, radcliffe, moya, federer, clijsters	Tennis and Athletics
2	music, band, album, chart, singer, pop, rock, elvis, console, concert	Music and Pop Culture
3	broadband, bt, mci, mobile, user, conte, verizon, lg, provider, plane	Telecom and Internet Services
4	labour, tory, ukip, lord, kennedy, lib, immigration, democrat, dems, blunkett	UK Politics and Immigration
5	film, oscar, actress, theatre, aviator, festival, starring, comedy, hollywood, nomination	Film and Entertainment
6	virus, software, google, blog, spam, patent, spyware, id, directive, animal	Technology and Cybersecurity
7	gadget, mobile, handset, camera, multimedia, ce, portable, henson, laptop, ruddock	Consumer Electronics and Gadgets

Table 5
Topic words and inferred categories for TripAdvisor dataset (num topics = 8).

#Topic	Top words	Inferred category
0	hotel, room, location, great, good, staff, breakfast, clean, stay, helpful	General Hotel Experience
1	sirenis, valley, rincon, argonaut, ramada, casitas, occidental, colonial, lti, balinese	Resort and Hotel Names
2	rincon, sirenis, valley, argonaut, ramada, occidental, lti, roach, palladium, colonial	Popular Hotel Chains
3	argonaut, ramada, valley, roach, stating, filthy, pet, confirmation, receipt, signature	Hotel Complaints and Cleanliness Issues
4	argonaut, valley, ramada, pet, signature, vieques, patron, raffle, casitas, skyline	Resort Booking and Pet-Friendly Travel
5	ramada, valley, argonaut, roach, signature, patron, stating, trash, emergency, promise	Negative Reviews and Service Issues
6	argonaut, ramada, valley, roach, pet, starwood, signature, bucket, dumont, midtown	Urban Hotels and Maintenance
7	argonaut, valley, sirenis, ramada, stating, pan, vacationer, benjamin, roach, signature	Mixed Feedback and Guest Experience

The distribution of maximum topic probabilities across the BBC News dataset is illustrated, based on the QTopic model with 8 latent topics (Fig. 5). The histogram indicates that most documents have a maximum topic probability in the range of 0.16 to 0.18. It can be concluded that documents are evenly distributed across topics, rather than being strongly aligned with a single topic.

Moreover, the topic assignment distribution is also presented to illustrate the distribution of documents across the identified topics (Fig. 6). The near-uniform distribution across all 8 topics demonstrates that QTopic assigns topics in a balanced manner, with no single topic dominating the corpus, supporting diversity and even topic usage.

A visual analysis of topic structure is presented by an intertopic distance map on the BBC News dataset (Fig. 7). This map is based on the Jaccard similarity of top-10 words per topic, considering 8 topics. The clear separation between points indicates that the topics are well-separated and semantically distinct, with minimal word overlap.

A t-SNE plot of document embeddings is also presented in the same setting, where documents are represented in a reduced 2D space based on their topic distributions (Fig. 8). The color-coded clusters correspond to different topics, representing coherent groups of documents in topic space. The well-defined and non-overlapping clusters suggest that QTopic is effective in capturing discriminative topic structure, leading to meaningful and well-defined document clusters.

5.2. Coherence and diversity analysis

We employed three different quantitative metrics—coherence C_V, topic diversity, and Jaccard similarity— to assess the quality of the topics discovered. These metrics collectively measure the semantic consistency, uniqueness, and overlap of the generated topics. The QTopic model is compared to three baseline models: LDA, BERTopic, and Top2Vec.

Table 6 presents the comparative analysis of topic modeling performance across 4, 8, and 16 topics on the 20 Newsgroups dataset. QTopic achieves the best coherence (0.6446), perfect diversity (1.00), and zero overlap (JS = 0.00) at 4 topics, reflecting highly distinct and coherent topics. Its performance remains strong at 8 topics with maximum coherence (0.585) and diversity (0.988), but degrades at 16 topics with the increasing number of qubits. BERTopic yields the highest coherence at 16 topics (0.689); however, high diversity and low JS are achieved by the Top2Vec model. While traditional LDA demonstrates better coherence as the number of topics increases, it tends to produce more overlapping topics.

The performance of LDA, BERTopic, Top2Vec, and QTopic on the BBC News dataset with 4, 8, and 16 topics is summarized in Table 7. QTopic outperforms all baselines at 4 topics, achieving the highest coherence (0.619), maximum diversity (1.00), and no topic overlap. At 8 topics, BERTopic achieves the highest coherence (0.676), while QTopic produces diverse and distinct topics. However, QTopic's performance declines at 16 topics, where BERTopic maintains high coherence and Top2Vec achieves strong topic diversity.

Topic discovery results on the TripAdvisor dataset are summarized in Table 8. Consistent with earlier datasets, QTopic outperforms all baseline models at 4 topics, achieving the highest coherence score (0.495), diversity (0.850), and Jaccard similarity (0.071), indicating clear topic separation. At 8 topics, LDA discovers the most diverse topics (0.600), though coherence remains relatively low (0.463). QTopic, in contrast, achieves the highest coherence at 8 topics (0.499), balancing diversity and overlap effectively. With 16 topics, BERTopic reaches the best coherence (0.449), LDA maintains the highest diversity (0.581), and QTopic attains the lowest Jaccard similarity (0.068), highlighting minimal topic redundancy.

We compare our proposed QTopic model against several established classical topic modeling baselines and evaluate performance using topic coherence, diversity, and Jaccard similarity metrics. These classical models are widely adopted in prior research for extracting latent topics and serve as standard benchmarks [12,14]. Experimental results demonstrate that the quantum-enhanced QTopic framework consistently outperforms classical models by generating more coherent and diverse topics, particularly when the number of topics (mapped to qubits) is relatively small. This advantage represents the expressive power of quantum

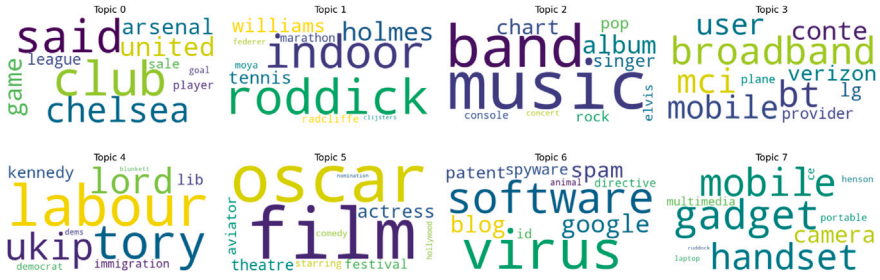


Fig. 3. Topic word clouds generated from the top-10 words per topic using QTopic on the BBC news dataset.

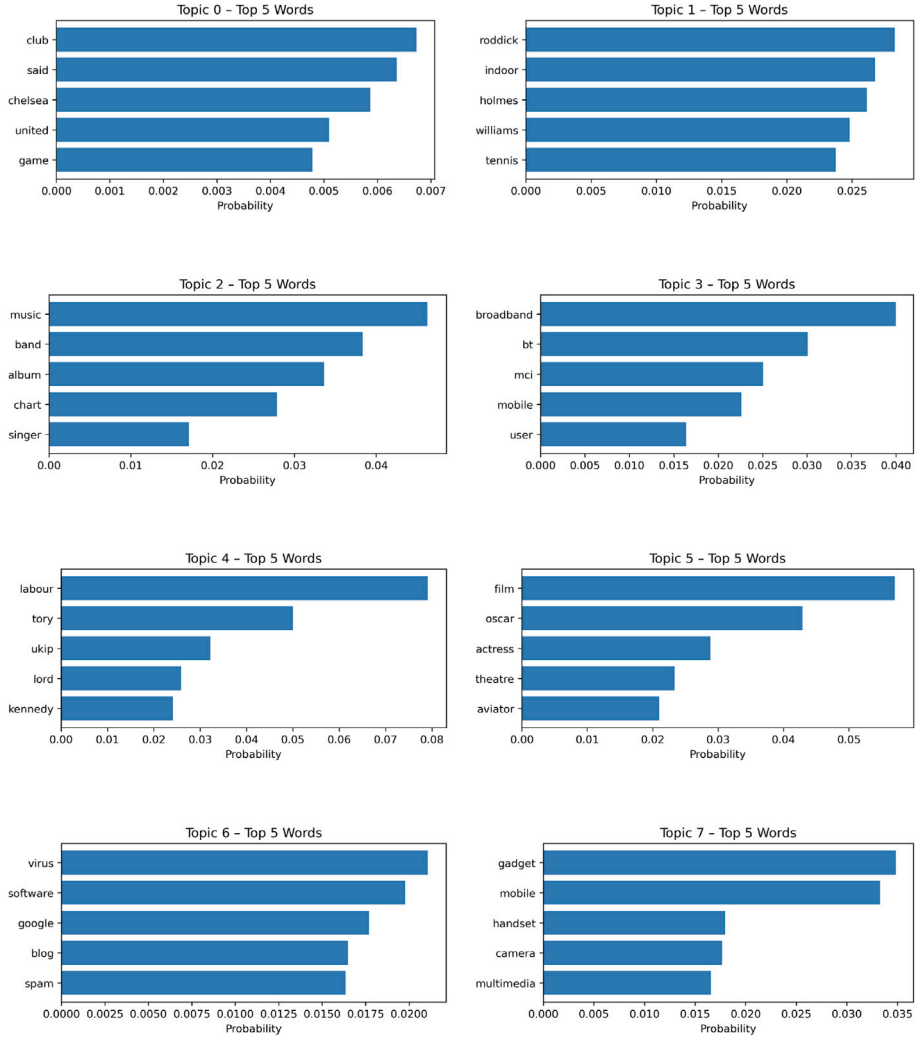


Fig. 4. Top words for each topic discovered by QTopic on the BBC news dataset.

circuits, which leverage quantum properties to explore universal and relevant topics.

However, with the increase in qubit number as well as the topics, the model begins to face significant limitations in hardware. Current noisy intermediate-scale quantum (NISQ) devices support only a limited number of stable qubits, making it challenging to scale QTopic to larger topic counts. Despite this, QTopic still demonstrates promising performance under realistic quantum hardware constraints, showing that hybrid quantum-classical models can provide meaningful benefits even in near-term settings.

5.3. Ablation study

This section presents an ablation study to evaluate the impact of individual components within the proposed framework. The analysis considers quantum sampling, PCA-based dimensionality reduction, and diversity regularization, compared against the proposed QTopic model. A limited number of topics are used to emphasize the contributions of quantum and other components. The experimental results with 4 topics on the 20 Newsgroups, BBC, and TripAdvisor datasets are summarized in Table 9.

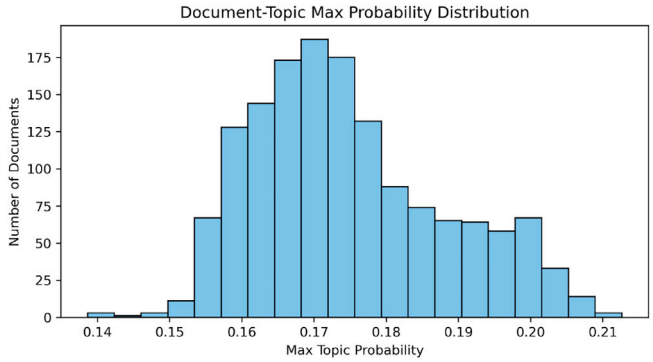


Fig. 5. Distribution of maximum topic probabilities per document.

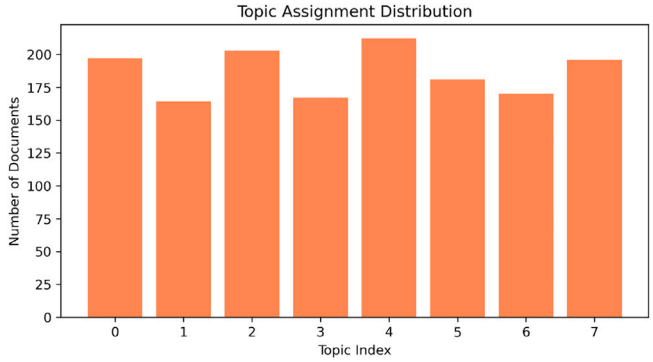


Fig. 6. Topic assignment distribution across topics.

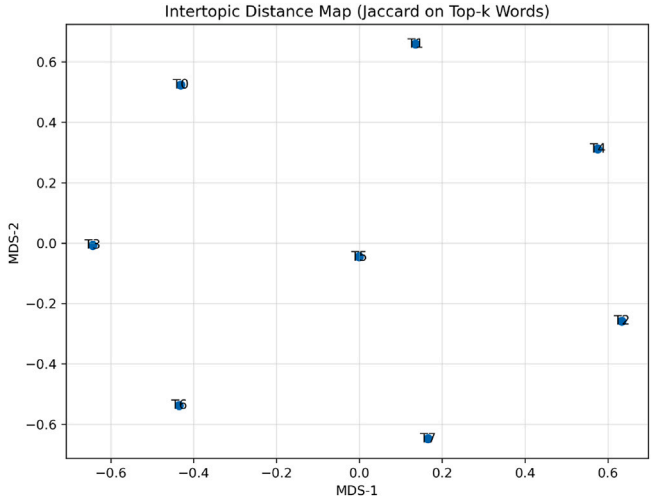


Fig. 7. Intertopic jaccard distance map over top-10 words.

The ablation results highlight the contribution of each component in the proposed framework. QTopic consistently achieves the highest coherence and topic separation, with C_V scores on 20 Newsgroups (0.645), BBC News (0.619), and TripAdvisor (0.495). The removal of the quantum module lowers coherence on 20 Newsgroups (0.436), BBC News (0.543), and TripAdvisor (0.397), which indicates that quantum sampling is crucial for topic structure and separation. PCA-based dimensionality reduction is more effective on larger datasets, where it stabilizes learning and maintains compact representations. Removal of the diversity penalty reduces topic distinctiveness, resulting in lower

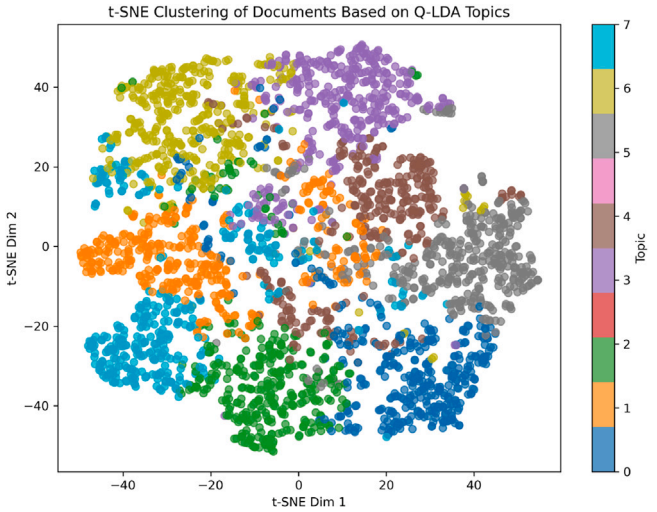


Fig. 8. t-SNE plot of document clusters based on QTopic model topic representations.

Table 6
Comparison of topic models with different topic counts on 20 Newsgroups.

Model	4 Topics			8 Topics			16 Topics		
	CV	TD	JS	CV	TD	JS	CV	TD	JS
LDA	0.501	0.725	0.159	0.539	0.750	0.074	0.600	0.775	0.033
BERTopic	0.612	0.967	0.018	0.567	0.886	0.035	**0.689**	0.853	0.031
Top2Vec	0.547	**1.00**	**0.00**	0.434	0.883	0.028	0.522	**0.938**	**0.006**
QTopic	**0.645**	**1.00**	**0.00**	**0.585**	**0.988**	**0.002**	0.466	0.544	0.06

Table 7
Comparison of topic models with different topic counts on BBC News.

Model	4 Topics			8 Topics			16 Topics		
	CV	TD	JS	CV	TD	JS	CV	TD	JS
LDA	0.496	0.750	0.146	0.511	0.650	0.126	0.503	0.619	0.104
BERTopic	0.539	0.867	0.094	**0.676**	0.871	0.051	**0.651**	0.887	0.018
Top2Vec	0.526	**1.00**	**0.00**	0.516	0.933	0.019	0.526	**0.914**	0.022
QTopic	**0.619**	**1.00**	**0.00**	0.542	**0.988**	**0.002**	0.462	0.888	**0.01**

Table 8
Comparison of topic models with different topic counts on TripAdvisor.

Model	4 Topics			8 Topics			16 Topics		
	CV	TD	JS	CV	TD	JS	CV	TD	JS
LDA	0.424	0.675	0.182	0.463	**0.60**	**0.128**	0.437	**0.581**	0.084
BERTopic	0.442	0.767	0.138	0.460	0.557	0.184	**0.449**	0.547	0.131
Top2Vec	0.463	0.750	0.128	0.363	0.513	0.174	0.354	0.463	0.149
QTopic	**0.495**	**0.850**	**0.071**	**0.499**	0.525	0.235	0.356	0.475	**0.068**

Table 9
Ablation study results on 20 Newsgroups, BBC News, and TripAdvisor datasets.

Model Variant	20 Newsgroups			BBC news			TripAdvisor		
	CV	TD	JS	CV	TD	JS	CV	TD	JS
Full QTopic	**0.645**	**1.00**	**0.00**	**0.619**	**1.00**	**0.00**	**0.495**	**0.850**	**0.071**
QTopic w/o PCA	0.533	0.675	0.217	0.475	0.950	0.019	0.329	0.725	0.153
QTopic w/o Quantum	0.436	0.825	0.080	0.543	**1.00**	**0.00**	0.397	0.750	0.119
QTopic w/o Diversity	0.452	0.825	0.071	0.397	0.925	0.029	0.312	0.650	0.251

TD values (BBC News 0.925, TripAdvisor 0.650) and higher JS, demonstrating its role in preventing topic overlap. Overall, the combination of quantum sampling, PCA, and diversity regularization achieves the best balance between coherence, diversity, and minimal redundancy.

6. Conclusions and future work

This study presents QTopic, a hybrid quantum-classical topic modeling framework that integrates parameterized quantum circuits into the traditional pipeline. Unlike classical topic inference, our method leverages quantum encoding, entanglement, and measurement to capture richer topic dependencies and enhance semantic separation. By replacing the variational sampling step with quantum circuit inference, QTopic introduces a novel perspective on modeling document–topic distributions through quantum-enhanced computation.

Experimental results demonstrate that QTopic achieves competitive results, often outperforming classical models across multiple evaluation metrics, including topic coherence, diversity, and inter-topic dissimilarity. In particular, the model shows improvements in maintaining balanced topic assignments and better disentangling overlapping topics, especially in lower topic-count settings. The framework is lightweight, interpretable, and hardware-efficient, demonstrating that hybrid quantum-classical methods can provide meaningful benefits even under near-term NISQ constraints.

Future research will explore variational training of quantum circuits, document-conditioned quantum encoding, and adaptive entanglement structures to enhance scalability and representational capacity. We also plan to extend QTopic to multilingual and domain-specific corpora to evaluate generalizability across diverse linguistic contexts. As quantum hardware continues to mature, larger qubit counts, improved coherence times, and deeper circuits will allow QTopic to scale to higher topic numbers. This will enable the capture of more complex topic dependencies, supporting richer latent representations and improved topic disentanglement.

In the long term, integrating hybrid variational optimization and quantum error–mitigation techniques could transform QTopic into a fully trainable and scalable framework suitable for large-scale real-world topic discovery. Such advances would support applications in complex NLP tasks, including cross-lingual topic analysis, domain adaptation, and dynamic topic tracking. Overall, QTopic highlights the promise of hybrid quantum-classical approaches to advance interpretable and expressive models for topic modeling and the broader field of quantum natural language processing.

CRediT authorship contribution statement

Monika Kabir: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammed Kaosar:** Writing – review & editing, Supervision, Methodology. **Ferdous Sohel:** Writing – original draft, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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